

Zoe Rand

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Employment

School of Marine and Environmental Affairs | University of Washington

Postdoctoral Scholar

Seattle, WA

2026 – Present

Education

University of Washington

Ph.D. Quantitative Ecology and Resource Management

Dissertation: Modelling the effects of whaling on blue whales and other baleen whales

Advisor: Trevor A. Branch

Seattle, WA

2020 – 2026

University of St Andrews

M.S. Marine Mammal Science

Thesis: Effects of duty cycling in passive acoustic monitoring of cetaceans

Advisor: Julie N. Oswald

St Andrews, Scotland

2019 – 2020

Oregon State University

Post-baccalaureate coursework, Fisheries and Wildlife Science

Corvallis, OR

2018 – 2019

Mount Holyoke College

B. A. Latin American Studies, Music

Summa Cum Laude

South Hadley, MA

2012 – 2016

Publications in review (2 first-author)

3. Trickey, J. S., Holt, M. M., **Rand, Z. R.**, Brewer, A. M., Emmons, C. K., Hanson, B. M. *In review*. The prevalence of ultrasonic antifouling sounds in the Salish Sea: High-frequency noise pollution in Southern Resident killer whale (*Orcinus orca ater*) critical habitat. *Marine Environmental Research*.
2. **Rand, Z. R.**, Branch, T. A., Punt, A. E., Maeda, H. *In review*. Estimates of annual survival for pygmy blue whales using historical data. *Marine Mammal Science*.
1. **Rand, Z. R.**, Branch, T. A., Punt, A. E., Maeda, H., Mizroch, S. A. *In review*. Estimates of blue whale growth and reproductive parameters from historical whaling data. *Marine Mammal Science*.

Peer-reviewed publications (6 first-author)

10. **Rand Z. R.**, Olson, P.A., Kinzey, D., Double, M. C., Miller, B. S., Matsuoka, K., Findlay K., Branch T. A. (2026). Critically Endangered Antarctic blue whales are increasing but still far from pre-whaling levels. *Endangered Species Research*. In press.
<https://doi.org/10.3354/esr01557>
9. **Rand Z. R.**, Koehn L. E., Morrigan, A. and Hanson, B. M. (2026). Increasing presence of Bigg's killer whales and changing seasonality of Southern Resident killer whales in Washington waters. *PLOS One* 21(6): e0350181.
<https://doi.org/10.1371/journal.pone.0350181>
8. **Rand Z. R.**, Branch, T. A. and Converse, S. J. (2025). Battle of the sexes: longer rorqual whale mothers have more female offspring. *Proceedings of the Royal Society B*. 292: 20251437. <https://doi.org/10.1098/rspb.2025.1437>
7. Branch, T.A., Monnahan, C.C., Leroy, E.C., ..., **Rand Z. R.**, et al. (2025). Separating historical catches among pygmy blue whale populations with the aid of recent song detections. *Marine Mammal Science*. e70003. <https://doi.org/10.1111/mms.70003>
6. **Rand Z.R.**, Branch, T. A. and Jackson J. A. (2024). High historical movement rates of Antarctic blue whales (*Balaenoptera musculus intermedia*) on Southern Ocean feeding grounds estimated from Discovery marks. *Endangered Species Research*. 55: 109-128.
<https://doi.org/10.3354/esr01361>
5. **Rand, Z. R.**, Ward, E. J., Zamon, J. E., Good, T. P., and Harvey, C. J. (2024). Using hidden Markov models to develop ecosystem indicators from non-stationary time series. *Ecological Modelling*, 495:110800. <https://doi.org/10.1016/j.ecolmodel.2024.110800>
4. Battle L., Patil A., Branch T.A., **Rand Z. R.** (2023) Visualizing historical whaling voyages over time. *Interactions* 30:22-23 <https://doi.org/10.1145/3611650>
3. Patil A., **Rand, Z.R.**, Branch, T.A., and Battle, L. (2023). WhaleVis: Visualizing the history of commercial whaling. *IEEE Visualization and Analytics (VIS)*, Melbourne, Australia, 96-100. <https://doi.org/10.1109/VIS54172.2023.00028>
2. **Rand Z.R.**, Wood J., Oswald J. (2022). Effects of duty cycles on the passive acoustic monitoring of Southern Resident killer whale (*Orcinus orca*) occurrence and behavior. *Journal of the Acoustical Society of America*, 151(3): 1651-1660.
<https://doi.org/10.1121/10.0009752>
1. Haver S., **Rand Z.R.**, Hatch L. et al. (2020) Seasonal trends and primary contributors to the low frequency soundscape of the Cordell Bank National Marine Sanctuary. *Journal of the Acoustical Society of America*, 148(2):845-858. <https://doi.org/10.1121/10.0001726>

Assessment documents (4 first-author)

6. **Rand Z. R.**, Branch, T. A. (2024). Updated historical catch series for Antarctic blue whales. *IWC Paper SC/69B/IA/01*.
5. **Rand Z. R.**, Branch, T. A. (2024). Preliminary results from an updated population assessment of Antarctic blue whales. *IWC Paper SC/69B/IA/05*.
4. Branch, T.A., Monnahan, C.C., Leroy, E.C., Shabangu, ..., **Rand, Z.R.**, et al., (2023). Further revisions to the historical catch separation of pygmy blue whale populations using contemporary song detections. *IWC Paper SC/69A/SH/09*.
3. Patil, A., **Rand, Z.R.**, Branch, T.A., Battle, L., (2023). WhaleVis: A new visualization tool for the IWC catch database. *IWC Paper SC/69A/GDR/04*.
2. **Rand, Z.R.**, Branch, T.A., (2023). Fetal sex misidentification and adaptive sex ratio behavior in large whales. *IWC Paper SC/69A/SH/06*.
1. **Rand Z.R.**, Branch, T., Jackson, J. (2022) Movement rates of Antarctic blue whales from Discovery marks. *IWC Paper 68D/SH/13*.

Funded research support

2026 – 2029	Washington Research Foundation	\$359,000
	Fellowship	
2024	University of Washington Graduate School	\$500
	Travel Award	
2024	University of Washington	\$1,500
	College of the Environment	
	Travel Award	
2023 – 2026	NMFS-Sea Grant	\$174,670
	Population and Ecosystem Dynamics	
	Fellowship	
2021 – 2023	University of Washington	\$1,400
	School of Aquatic and Fisheries Science	
	Travel Award	
2020-2021	University of Washington	\$31,000
	Quantitative Ecology and Resource Management	
	Fellowship	
2019	University of Washington	\$56,000
	R&A Ransome	
	Scholarship	

Honors and awards

- 2025 **Graduate Dean's Medalist**
College of the Environment, University of Washington
Recognizing excellence in academic achievement, leadership, and service
- 2024 **Best student presentation from North America**
25th Biennial Conference on the Biology of Marine Mammals, Perth, Australia
- 2024 **Best student poster presentation**
Pacific Seabird Group Annual Meeting, Seattle, WA

Committee membership

- 2025 – Present **Antarctica's Lost Giants Working Group**
Antarctic Insync
- 2023 – Present **Invited participant, Scientific Committee**
International Whaling Commission

Mentorship

Undergraduate mentees

Role	Name	Project Title	Start	Completion
Supervisor	Ashley Rendon	Antarctic blue whale fetal and calf growth rates from historical data	Jan. 2024	Dec. 2024

Training

- 2023 – 2024 **Identity Belonging and Inquiry in Science**
Mentorship training sessions focused on enhancing communication skills, adopting mentorship best practices, and integrating diversity, equity, and inclusion principles into mentorship approaches

Teaching

- 2022 – 2023 **Reader/Grader**
QSCI 458 Advanced Ecological Modeling
University of Washington
Delivered timely and helpful feedback to ~15 students each quarter on written assignments and code in R
- 2018 – 2019 **Educator**
Oregon Coast Aquarium
Engaged guests and community stakeholders with entertaining and informational activities that taught principles of biology and ecology, inspired

- love for ocean life and fostered understanding about the need for ocean conservation.
- 2018 – 2019 **Undergraduate Writing Consultant**
Oregon State University
 Taught writing skills one-on-one and in small groups with undergraduate and graduate students, including all aspects of the writing process and general strategies to succeed in college academic writing.
- 2013 – 2016 **Speaking Arguing and Writing Mentor**
Mount Holyoke College
 Taught strategies such as brainstorming, outlining, and revising, while working with students one-on-one and led writing and public speaking workshops for large groups of students. Collaborated with professors to support the writing progress of their students in first-year seminars.

Recent guest lectures

Year	Lecture topic	Course	Level	# of students
2026 & 2025	Whaling and the International Whaling Commission	Conservation and Management of Aquatic Resources	Undergraduate	90
2025	Population effects of whaling and estimating marine mammal abundance (Lecture + lab)	Evolution and Ecology of Marine Mammals	Undergraduate	12
2024	Plotting techniques using ggplot2 (Lecture + lab)	Beautiful Graphics in R	Graduate	25
2024	Adaptive sex ratio behavior in marine mammals (Lecture + lab)	Evolution and Ecology of Marine Mammals	Undergraduate	25

Facilitated workshops

- 2023 **Writing personal statements**
College of the Environment, University of Washington
 This workshop introduced undergraduate students to the process of writing personal statements for fellowships and graduate school and gave them hands-on practice with writing their own personal statements. 25 participants
- 2021 **Using GitHub for research**
School of Aquatic and Fisheries Science, University of Washington
 This workshop introduced GitHub and version control and provided hands-on experience with using it for ecological research. 20 participants

Presentations

8. **Rand Z. R.**, Branch, T., (2025). Bayesian models reveal endangered Antarctic blue whale population is increasing but still far from pre-whaling levels. Seminar Presentation. *SAFS Quantitative Seminar*, University of Washington, Seattle, WA. February 21, 2025.
7. **Rand Z. R.**, Koehn, L., Morrigan, A., Hanson, B. (2024). Spatiotemporal modeling of killer whale presence in the Salish Sea. Workshop presentation. *Orca Occupancy Indicator Workshop*, Seattle, WA. December 4, 2024.
6. **Rand Z. R.**, Branch, T., Converse, S. (2024). Longer mothers tend to have more female calves: evidence for adaptive sex ratio behavior in rorqual whales. Poster presentation. *25th Biennial Conference on the Biology of Marine Mammals*, Perth, WA, Australia.
5. **Rand Z. R.**, Ward, E. J., Zamon, J. E., Good, T. P., Harvey, C. J. (2024). Hidden Markov models identify regime shift from seabird at-sea-density in the Northern California Current. Poster presentation. *Pacific Seabird Group Annual Meeting*, Seattle, WA.
4. **Rand Z.R.**, Branch T., Jackson J. (2023) High movement rates of Antarctic blue whales (*Balaenoptera musculus intermedia*) on Southern Ocean feeding grounds estimated from historic mark-recovery data. Contributed oral presentation, *Ecological Society of America Annual Meeting*, Portland, OR.
3. **Rand Z.R.**, Branch T., Jackson J. (2022) Movement rates of Antarctic blue whales from historical data using Bayesian models. Seminar Presentation. *SAFS Quantitative Seminar*, University of Washington, Seattle, WA. October 21, 2022.
2. **Rand Z.R.**, Wood J. A., Oswald, J.N. (2022). Effects of Duty Cycles on the Passive Acoustic Monitoring of Southern Resident Killer Whale (*Orcinus orca*) Occurrence and Behavior. Poster presentation. *Society for Marine Mammalogy Biennial Conference*, Palm Beach, FL. August, 2022.
1. **Rand Z.R.** and Haver S. (2018). Identification of marine mammal species in the Cordell Bank National Marine Sanctuary using passive acoustic data. Conference Presentation. *Research and Advances in Fisheries, Wildlife and Ecology*, Corvallis, OR. April, 2018.

Technical skills

- **Population modeling:** integrated population models, assessments of endangered species, matrix population models, mark-recapture models, estimating demographic rates
- **Statistics:** Bayesian methods, maximum likelihood methods, generalized linear and generalized linear mixed models (GLM, GLMM), generalized additive models (GAM), species distribution models, hidden Markov models; model fitting, diagnostics, selection, validation, and predictive skills testing; simulation testing
- **Programming languages:** R (including *base*, *tidyverse*), Julia, Python, TMB (C++) Stan, JAGS, NIMBLE
- **GitHub workflow and version control.** Code for selected projects can be found at <https://github.com/zoer27>
- **Technical writing** and communicating to technical and non-technical audiences

Academic service

2023 – 2024	Search committee member Center for Quantitative Science (CQS) director University of Washington
2022 – 2024	Student advisory council member College of the Environment University of Washington
2022 – 2024	Curriculum committee member School of Aquatic and Fisheries Sciences University of Washington
2022 – 2023	Quantitative seminar organizer School of Aquatic and Fisheries Sciences University of Washington
2021 – 2022	Students Experience Aquatic Sciences board member and lesson committee lead University of Washington

Reviewer

Canadian Journal of Zoology | Marine Ecology Progress Series | PLOS One | Proceedings of the Royal Society B | Science Advances | Ecological Modelling

Society memberships

Society of Marine Mammalogy (since 2019)